

ENERGY OPTIMISATION FOR UAV-MOUNTED INTELLIGENT REFLECTING SURFACES

Deep Reinforcement Learning for Disaster Communications

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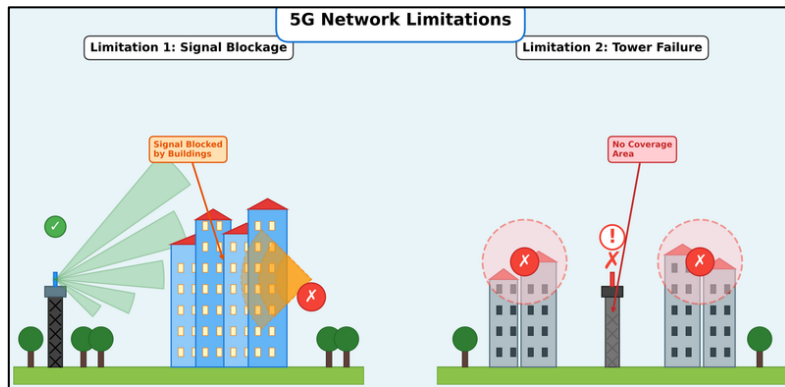
Problem Statement

Communication Challenges

- Disaster scenarios: **Infrastructure failure**
- Urban dead zones: **Signal blockages**
- Remote areas: **No coverage**

Technical Complexity

- **Non-convex optimization**
- **70-dimensional** action space
- Coupled variables: trajectory + phase shifts



Research Question: Can deep reinforcement learning jointly optimize UAV positioning and RIS phase shifts to maximize energy efficiency while maintaining user coverage?

Key Challenges & Solutions

Dynamic Channel Environment

Channel conditions constantly **vary** where each deployment has **unique channel conditions**

- ✓ **Utilise feedback systems to ensure the system is adaptable to maintain coverage**

Limited System Resources

Constraint by UAV fixed battery capacity

- ✓ **Optimise system parameters for energy efficiency**

Simulation Accuracy

Investigation is limited simulation experimentation.

- ✓ **Develop detailed channel models with relevant channel fading and environmental noise for relative comparison**

Non-Convex Optimisation

Optimisation of state/action space using traditional algorithms is slow and unsuitable for real-time applications

- ✓ **Utilise reinforcement learning algorithms to allow for fast system response**

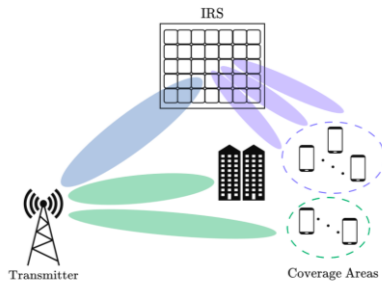
Proposed Solution



UAV

10 UAVs

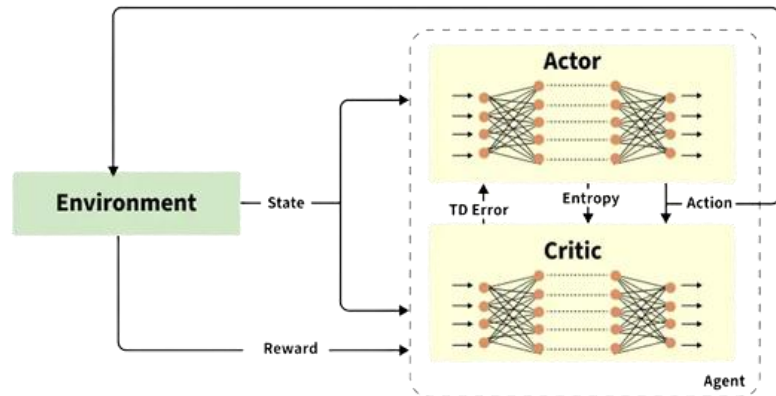
3D mobility
Rapid deployment



IRS

32-Element Arrays

Passive beamforming



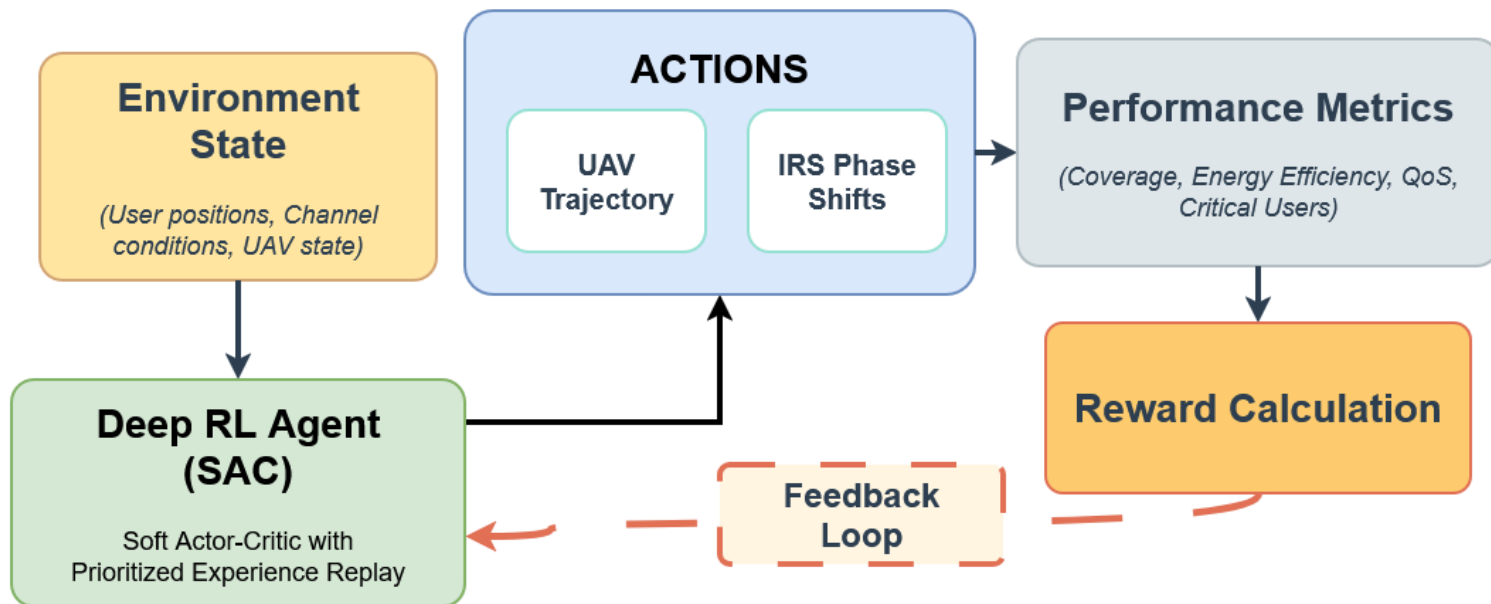
SAC

Deep RL

Real-time adaptation

System Architecture

Deep RL Optimization Framework



SAC Algorithm Architecture

State Space

- **148 dimensions**
- Channel, position, energy

Action Space

- **70 dimensions**
- Velocity + phase shifts

Network

- Twin critics: **512×512**
- Soft updates: $\tau=0.005$

Training

- **2000 episodes**
- Batch size: 256

Technical Approach

SYSTEM CONFIGURATION

UAV Fleet	Number $\times$ IRS Elements	10 UAVs $\times$ 32 elements
Coverage	Users / Area	50 users / 1 km ²
Flight Specs	Altitude / Speed / Battery	50-200m / 20 m/s / 1000 Wh

CHANNEL MODELING

Propagation	Rician K-factor / Path Loss α	10 dB / 2.2 (urban)
RF Specs	Frequency / Bandwidth	28 GHz / 100 MHz
Power	BS Tx / Noise Floor	30 dBm / -90 dBm
Coherence	Channel Coherence Time	100 ms

IMPLEMENTATION

RL Platform	Python / Stable Baselines3	SAC Agent
Computing	PyTorch / MATLAB / GPU	RTX 3090
Training	Episodes / Time / Dimensions	2000 / 3-4 hrs / 148D-70D

Key Contributions

Novel SAC Application

First for joint UAV-IRS optimization

1

Performance Gains

33% energy | **9% spectral** efficiency

2

Real-time Capability

0.50ms inference | **160× faster**

3

Energy Efficiency Performance

SAC Achievement

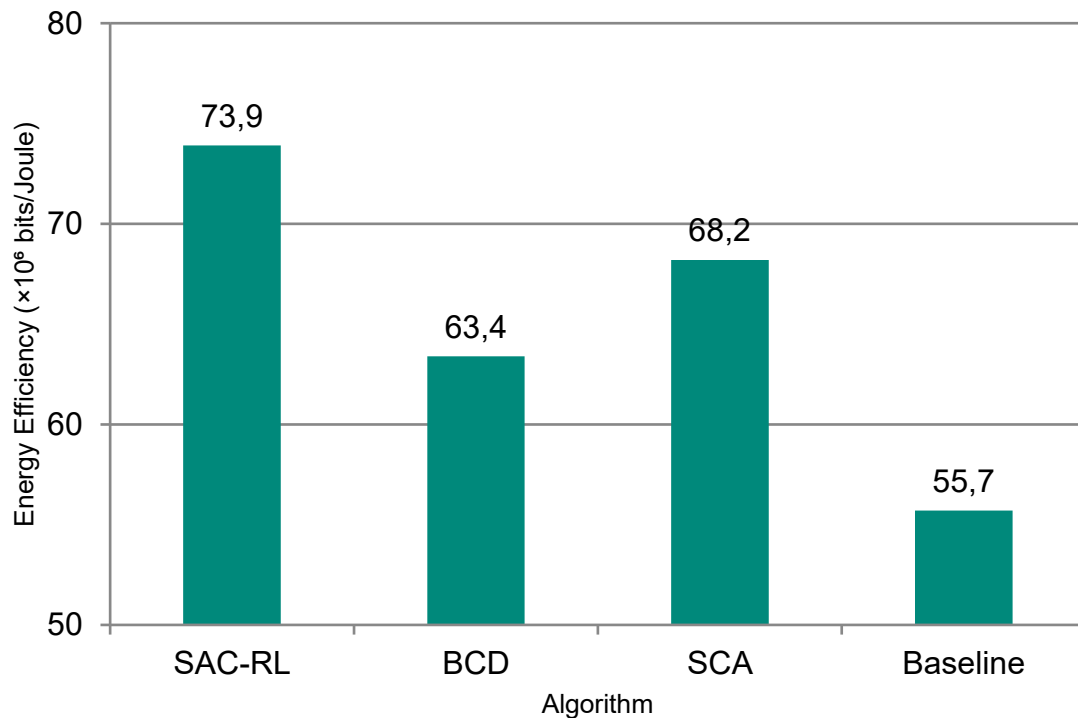
73.9 Mb/J

33% improvement over baseline

vs. Traditional Algorithms

+16.5%

over SCA method



Spectral Efficiency Gains

Throughput

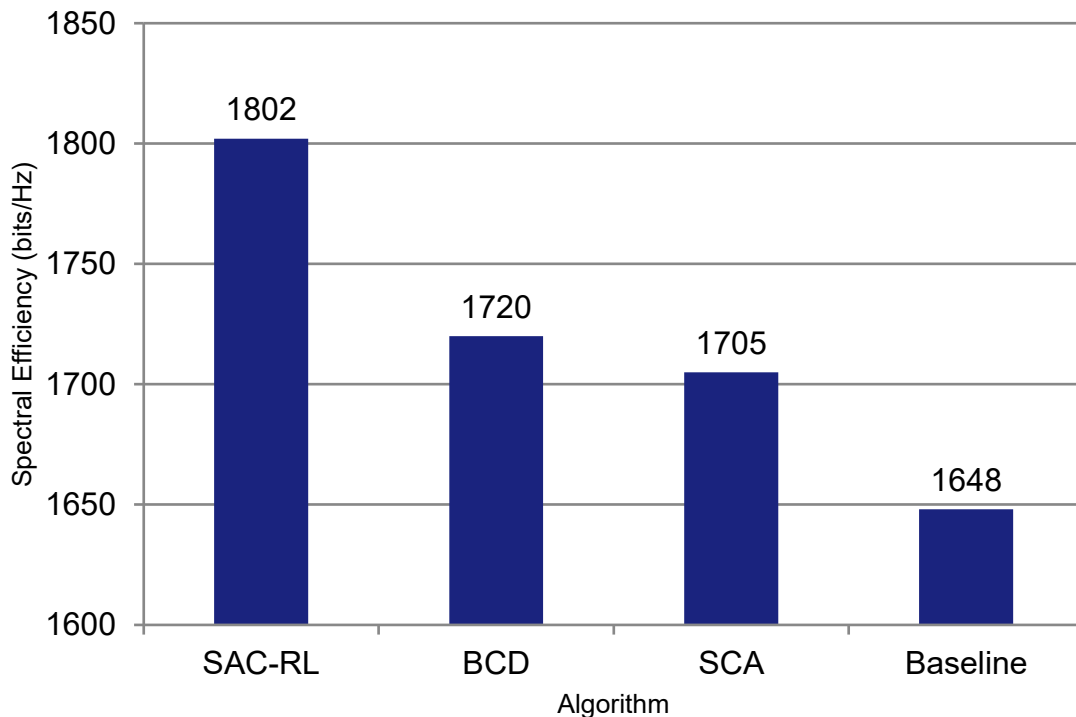
1802 bits/Hz

9% improvement over baseline

Energy Contribution

65%

50 ground users



Real-Time Performance

SAC Inference

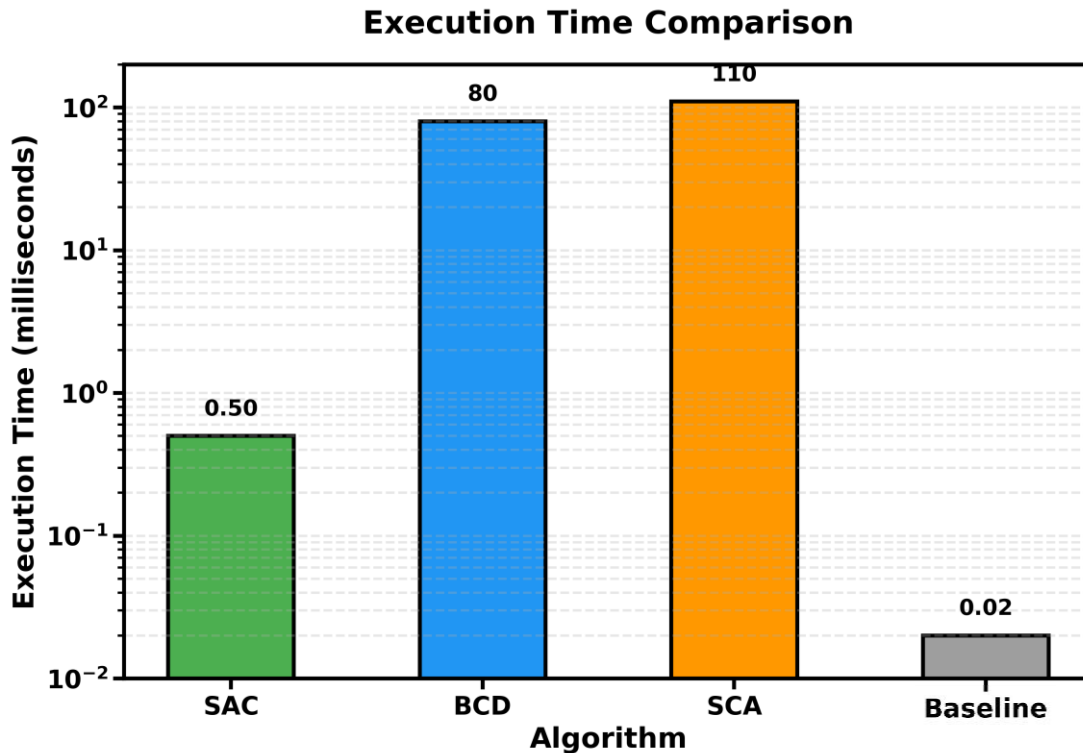
0.50 ms

Real-time capable

Speed Advantage

160× BCD

220× SCA



UAV Trajectory Optimization

Flight Path

40% shorter

Strategic hovering

Energy Impact

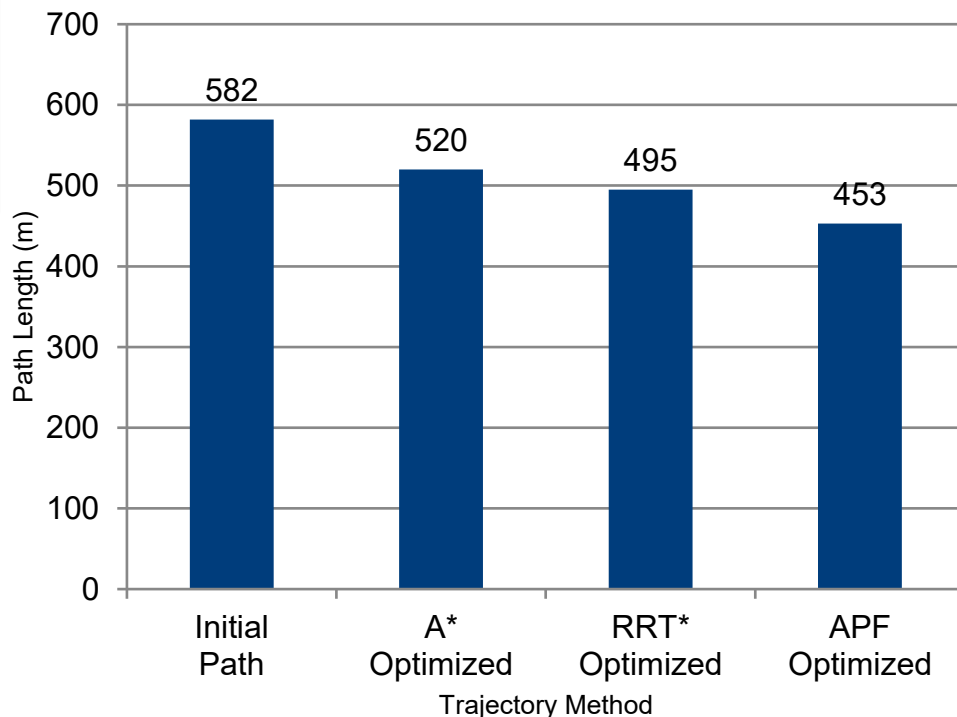
35% of gains

Trajectory contribution

Multi-Algorithm Approach

- A* pathfinding for obstacle avoidance
- RRT* for complex navigation
- Artificial Potential Field for smooth paths
- Differential Evolution optimisation

Result: 13-23% path efficiency improvement in dense obstacle scenarios



ML Method Comparison

Algorithm	Mean Reward	Std Dev	Inference (ms)	Conv. (Episodes)	Sample Efficiency	Training Stability	Final Perf (%)
SAC (Ours)	0.649	0.009	0.46	650	High	Excellent	90-98
PPO	0.644	0.136	0.22	10,000	Low	Medium	80-88
DDPG	0.815	0.002	0.17	1,500	Medium	Poor	75-85
DQN	0.819	0.004	0.35	3,700	Low	Medium	70-80
Random Forest	0.585	0.110	5.22	N/A	N/A	N/A	60-70
Gaussian Proc	0.806	0.001	0.20	N/A	N/A	Good	70-75
LSTM	0.412	0.245	383.09	5,000	Low	Poor	50-60

Key Finding: SAC achieves optimal trade-off between reward (0.649), stability ($\sigma=0.009$), and real-time performance (0.46ms). DDPG/DQN show higher raw rewards but converge to local optima. SAC: 160× faster than BCD, 220× faster than SCA.

Future Research Directions

5G MEC nodes enable **distributed SAC** training with **40% faster** convergence and sub-millisecond inference. This extends **battery life by 8-10%** through computational offloading and federated learning across swarms.

Edge Computing Integration

Multi-Agent SAC Framework

Graph neural networks coordinate 50+ UAVs with 10% state compression and collision avoidance. The framework achieves **scalability** through **decentralized** swarm coordination.

Strategic 10% active elements provide 20-30 dB gain **with 90% power reduction**. SAC-based mode switching doubles coverage while maintaining energy efficiency.

Hybrid Active-Passive IRS

Impact & Conclusions

Real-World Deployment

Flight time extended by 2 hours (2.7→4.3 hrs) enables extended emergency response operations

Scalability Tested

Tested with 2, 5, and 10 UAVs serving up to 50 users across 1km² disaster zones

Academic Contribution

First application of SAC to joint UAV-IRS optimisation in disaster scenarios

Industry Relevance

Direct integration to 5G/6G networks and emergency humanitarian operations

Key Takeaway

Deep RL successfully optimises complex multi-UAV systems, achieving 33% energy efficiency improvement while providing user coverage in simulated disaster scenarios

Individual Contributions

Mohemad Salman Kudia

- SAC agent architecture & training pipeline
- Neural network design & optimization
- Reward function engineering
- Training stability & convergence fixes
- Performance analysis & visualization

Yasteel Ramphal

- UAV-AIRS environment implementation
- Channel modeling & propagation physics
- Disaster scenario generation
- Baseline method implementations
- Comparative evaluation framework

Thank You

Questions & Discussion

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GitHub: github.com/yourusername/UAV-AIRS-RL